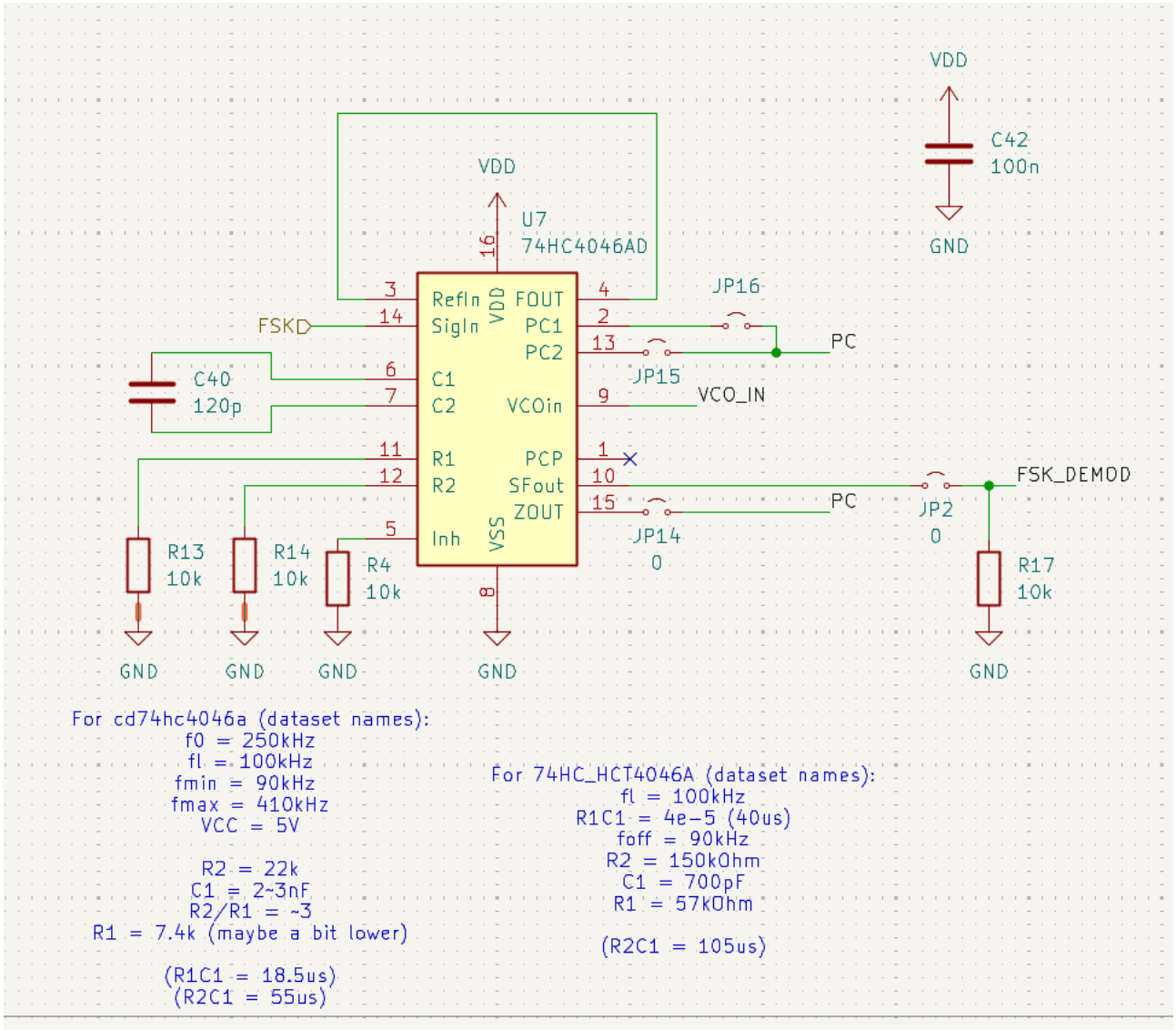
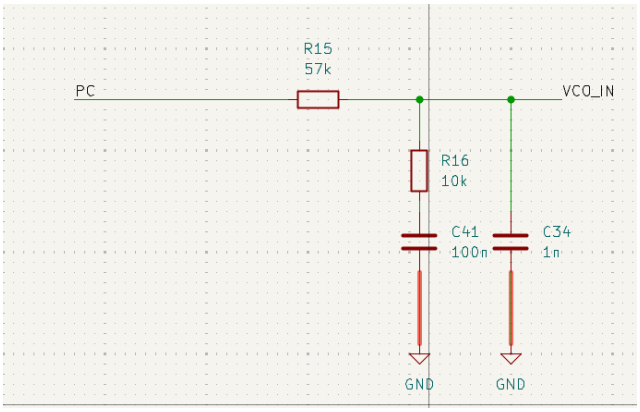


Available components close:
- CD74HC4046AE
- 22kOhm resistor
- 6.8kOhm resistor + small value in series (200 to 400)
- 1nF cap

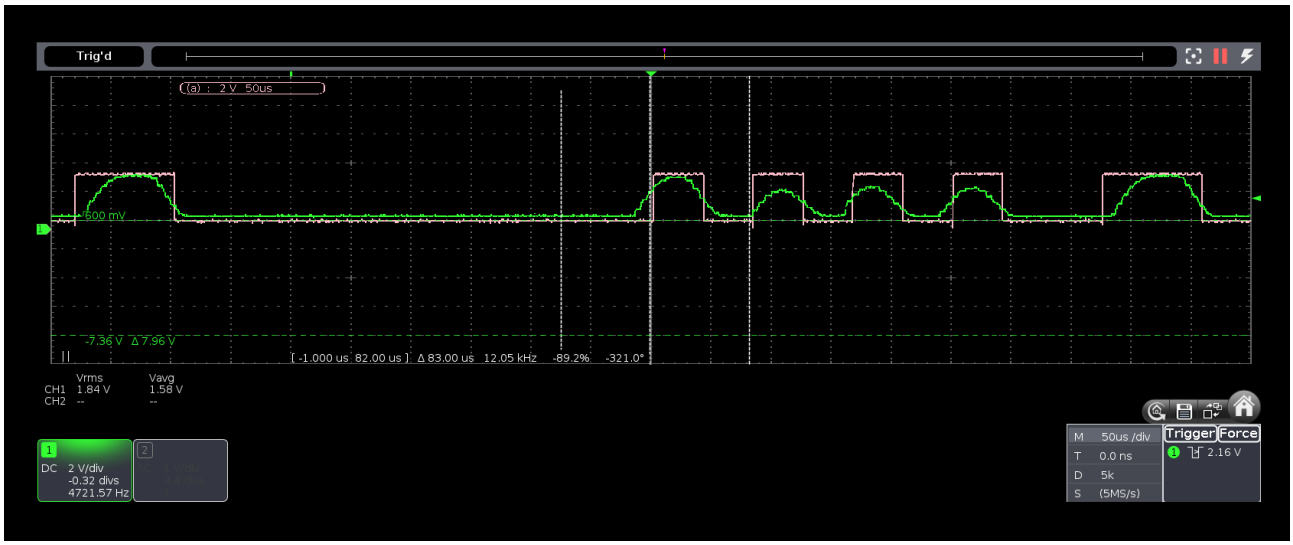
- Nearest to 57kOhm is 47kOhm followed by 10kOhm, but for testing 47kOhm is okay



In practice:
- With these values, offet frequency (feeding 2.3V) is ~400kHz, almost twice as expected, so we increase R2 to 47kOhm and C1 to 1nF, giving 237kHz central frequency (feeding 2.4V out of 4.8V)
- Lower tuning range of 0.8V is appropriate
- Higher tuning range of 4V is a bit too high, maybe reduce a bit in real transmitter design
- R1 is chosen to reduce fmax / fmin a bit, with 220kOhm it goes from 160kHz to 315kHz, which I pretty decently close to the wanted 150 to 350kHz4

in summary:
- R2 = 47kOhm
- C1 = 1nF
- R1 = 220kOhm (roughly R1/R2 = 5)

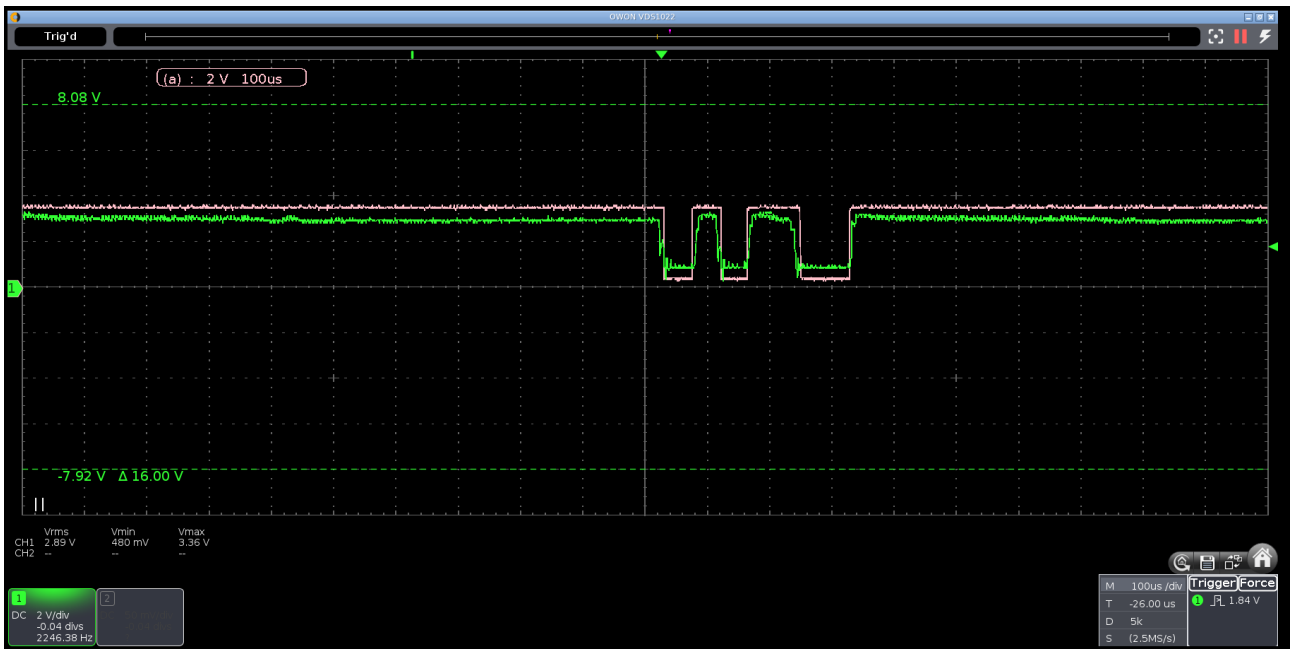
I think the modelling of the current phase detector is not correct for example the followign result is obtained by 10kOhm series resistor, 1nF to ground (no secondary path) as the loop filter, for the desired bitrate.



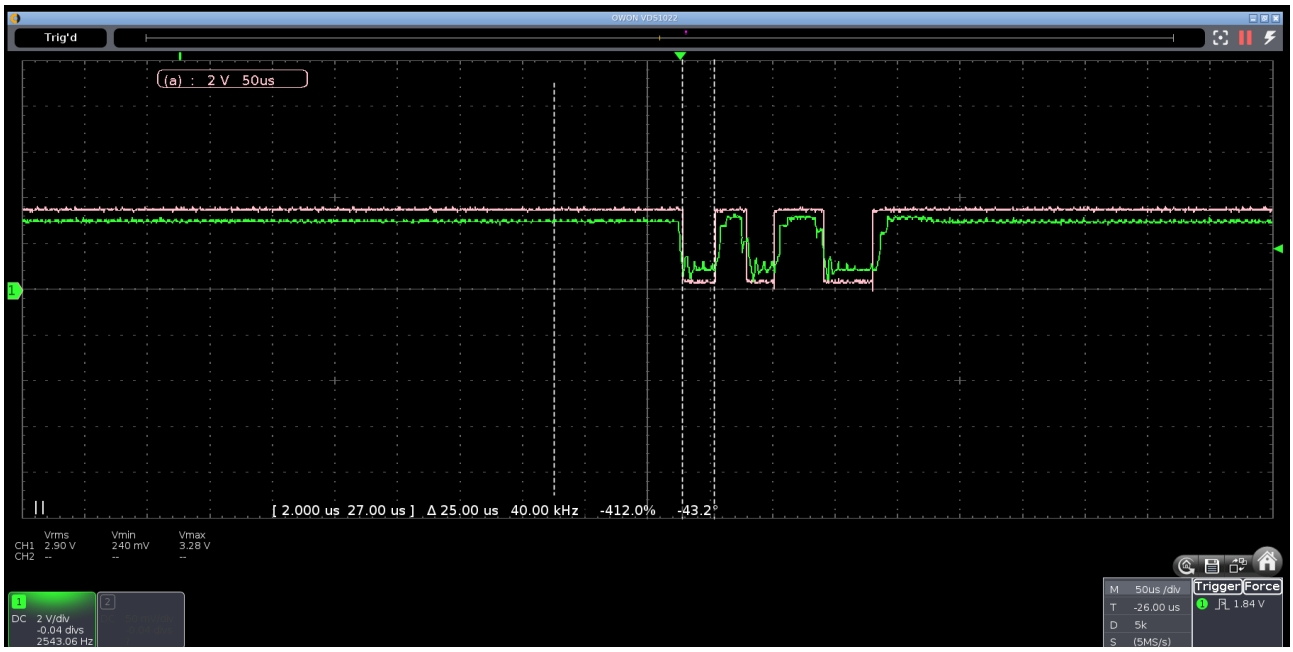
On the model, such a design oscillates wildly going into very negative regions. This is likely due to lack of simulation of the voltage limits (i.e. the phase comparator is not a true current source). TODO: simulate with current source as high / low voltage, and on high impedance just match the existing voltage on the node or similar technique? Assuming the cap doesn't swing noticeably in voltage is totally incorrect as evident by reality! This model may only work for very small signals near VCC/2.

Similarly, our loop filter modelling also considers the system to be marginally stable, so something's wrong on model. Because our filter allows this configuration to be used later on, going ahead with fabrication (in fact, this 10kOhm + 1nF would likely work just fine).

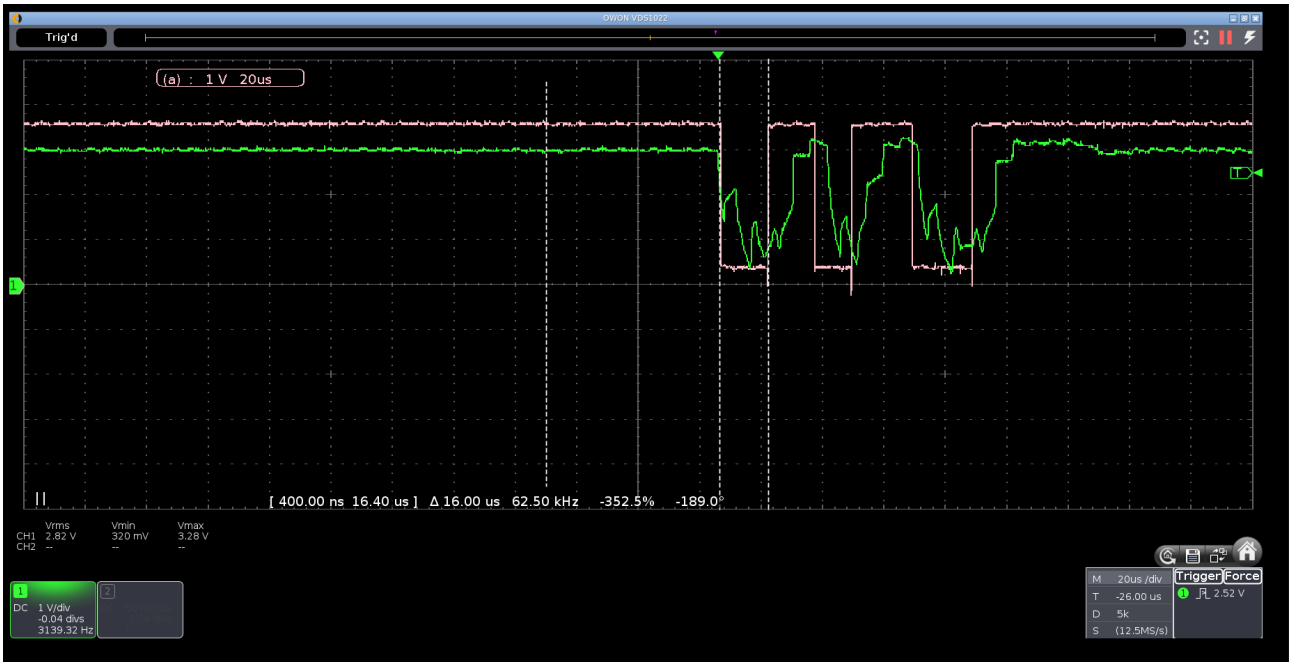
Tracking is seriously improved by
- Replacing the 10kOhm by a 470ohm resistor
- Adding a 10kOhm in series with 100nF to ground:



With this the post loop-filter is seriously improved. With this it can even work reliably at 40kHz clocking:



And could possibly work at 62kHz clock rate with some post-filtering or a strong hysteretic comparator:



Input to attenuator U46, AD603, is 240mV p2p
Likely U13 (LMH6551) is fried or something that feeds it is fried, see schematic.

Conclusion: replace the LMH6551.